

Postulates of Quantum Mechanics

Postulate 1

The complete physical state of a quantum system at any given time is described a single-valued, continuous, and square-integrable mathematical function known as the state function or the wave-function.

All properties of a quantum-mechanical-system are defined by $\psi(r, t)$

↳ wave-function

$$p(x) = \psi^*(x) \cdot \psi(x) \\ = |\psi(x)|^2$$

$$\int_{-\infty}^{\infty} p(x) \cdot dx = 1$$

Postulate 2

Every physically measurable property (called an observable) is represented by a specific mathematical operator in Quantum Mechanics (linear or Hermitian operator)

$$x \rightarrow \hat{x} = x$$

$$p \rightarrow \hat{p} = -i\hbar \nabla$$

$$V(x) \rightarrow \hat{V} = V(x)$$

$$T \text{ (K.E.)} \rightarrow \hat{T} = \frac{\hbar^2}{2m} \nabla^2$$

$$E \text{ (T.E.)} \rightarrow \hat{H} = \hat{T} + \hat{V} = \frac{\hbar^2}{2m} \nabla^2 + V(x)$$

$$\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Postulate 3

$$\hat{A}|\psi\rangle = a|\psi\rangle \rightarrow \text{Eigenvalue Equation}$$

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle \rightarrow \text{Prob. of measurement outcome } m$$

It's a collection of measurement operators $\rightarrow \{M_m\}$

$$\text{Posterior State} \rightarrow \frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}$$

$$\sum M_m^\dagger M_m = I \rightarrow \text{Completeness Relation}$$

$$m_0 = 100001$$

$$m_1 = 110011$$

$$p(0) = \frac{1}{\|m_0\|} = \frac{1}{\|\alpha\|} = \alpha$$

$$140 = \alpha 100 + \beta 110$$

$$\text{Posterior state} = \frac{m_0}{\|m_0\|} = \frac{\alpha 100}{|\alpha|}$$

$$p(1)$$

$$p(+)$$

$$p(-)$$

$$m_+ = 110011$$

$$m_- = 100011$$

Ans.

